Lab 3: Introduction to Microprocessor Interfacing

Goals

1. Understanding of the fundamental concepts of Assembly Language
2. Able to write Simple Assembly program using ARMv7 ISA to interface with simple IO
3. Develop familiarity with a IO system of a Computer System Simulator

Summary of Tasks:

1. Consult ARM Instruction Set Manual available at Canvas
2. Understand the following assembly example
    a. The meaning of each of the identifiers in the program
    b. How data segments and instruction segments work together
    c. Understand the IO addressing and how to use it
3. Implement the following two program:
    a. Write a program that will find that will print 0 in the seven segment display for invalid entry of a triangle's side such as side is zero or negative or sum of two sides is not greater than the third side
    b. Write another program that will calculate the area and perimeter of a rectangle and print in the 7-segment display. Take integer inputs for both sides of the rectangle of the rectangle.
4. Do Something cool. The following are a few examples.
    a. Calculate the area of a circle or a triangle
    b. Calculate the perimeter of a circle or a triangle

Assembly program example:

```
 .org 1000 @program start at memory address 1000
.equ ADDR_7SEG1, 0xFF200020
.equ ADDR_7SEG2, 0xFF200030
.equ X, 3
.equ Y, 4
.equ Z, 5

.global _start
_start:

 MOV R1, #X
 MOV R2, #Y
 MOV R3, #Z
 MUL R5, R1, R1
 MUL R6, R2, R2
 ADD R5, R6
```

```
 MUL R6, R3, R3
 CMP R5, R6
 BEQ _same
 ldr r0, =ADDR_7SEG1
 ldr r1, =#0b0111111 // bits 0b0111111 will display 0
 str r1, [r0]                // Write to 7-seg display
 mov r7, #1  @ syscall number for "exit"
 swi 0       @ initiate the syscall
_same:
 ldr r0, =ADDR_7SEG1
 ldr r1, =#0b0000110 // bits 0b0000110 will activate segments 1 and 2, display 1
 str r1, [r0]

   @ For example, to exit a program in ARM assembly, you can use the following:
mov r7, #1  @ syscall number for "exit"
swi 0       @ initiate the syscall

.end
```

Lab Demo
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Demo Part 3, Part 4 (optional) to lab instructor. And answer the following questions and submit a Canvas report

1. Every IO peripheral has addresses in the Altera DE-1 board, find addresses of all the LEDs, switches and push button?

2. Write a pseudo algorithm for ARM assembly that can calculate the area of a triangle when three lengths of three valid sides are provided.

3. Can we just load the content of 0xFF200040 and it will let us know which switch is ON? (Hint: read Altera DE1 manual to understand how the switches work)